Colin Barber

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SUMMARY

Systems-focused software engineer with hands-on experience in embedded and robotics platforms, maritime security research, and low-level C/C++ development. Skilled in debugging at the hardware-software boundary, network protocol analysis, and building reliable software for constrained, high-stakes environments. Security-minded by training, with a focus on software with adversarial conditions in mind from the start.

EDUCATION

Texas A&M University-Commerce
Bachelor of Science, Computer Science — *Cum Laude*

PROFESSIONAL EXPERIENCE

Computer Science Tutor | ClubZ Tutoring — Jul 2024 – Present

- Deliver personalized CS and Mathematics instruction to 100+ students, designing individualized learning plans from baseline assessments to accelerate progress.
- Translate complex topics, data structures, algorithms, and programming fundamentals into clear, practical exercises tailored to each learner

Project Manager & Lead Developer | ATproto Decentralized Web Client — Aug 2023 – May 2024

- Architected and delivered a full-stack microblogging platform built on PostgreSQL, Java, and custom decentralized identity (DID) protocols, implementing end-to-end security and self-sovereign identity management.
- Designed and owned the DID generation protocol, enabling users to authenticate without reliance on centralized identity providers — a core security differentiator of the platform.
- Applied test-driven development (TDD) to maintain feature alignment with user stories and enforced code quality through structured peer reviews.
- Led sprint planning and delivery using Jira and Agile methodologies across a multi-person team, balancing technical execution with project milestone accountability.

Cybersecurity Researcher | Cross The Border Threat Screening & Supply Chain Defense, a DHS Center of Excellence — May 2023 – Jul 2023

- Built test bed environments replicating NG911 emergency communications infrastructure and maritime vessel OT/SCADA systems, reconstructing them at sufficient fidelity that findings transferred to real-world deployments.
- Executed offensive security assessments against those environments to identify exploitable vulnerabilities and map attack paths, surfacing exposed Modbus/DNP3 endpoints.
- Contributed findings to published research on incident response improvements for next-generation 911 systems, translating test bed results into recommendations for emergency communications operators.
- Developed Python tooling to automate assessment and response workflows, reducing assessment cycle time by 40% and establishing repeatable procedures adopted by the research team.
- Built data visualization dashboards surfacing attack patterns across test runs, and documented segmentation and firewall hardening against the NIST Cybersecurity Framework.

Security Researcher | IStarLabs — Aug 2021 – Sep 2022

- Operated as part of a purple team, designing Zero-Trust network architecture while simultaneously stress-testing it through offensive security techniques to validate real-world resilience.
- Integrated Cloudflare, Tailscale, and OKTA to implement identity-aware access controls and enforce least-privilege across an enterprise network.
- Conducted network analysis, performance testing, and reverse engineering on IoT robotics platforms, identifying exploitable vulnerabilities and implementing targeted hardening measures.
- Developed a proof-of-concept kill chain modeled on the Mirai Botnet to map IoT attack surfaces and validate the effectiveness of implemented security enhancements.
- Maintained detailed technical documentation of architecture decisions, threat findings, and remediation steps for internal knowledge transfer.

CORE EXPERTISE

Languages	Python, C, C++, C#, Java, Go, Rust, Solidity, Ladder Logic
Security	Pentesting, Purple Team, Zero-Trust, IoT Security, Digital Forensics, Endpoint Security, Firewall & Vulnerability Management, Reverse Engineering, Network Engineering, NIST Framework
Development	Full-Stack, Decentralized Apps, REST APIs, Data Pipelines, Git, Jira, Confluence
Infrastructure	Docker, Kubernetes
Databases	PostgreSQL, MySQL, MongoDB, NoSQL, SQL